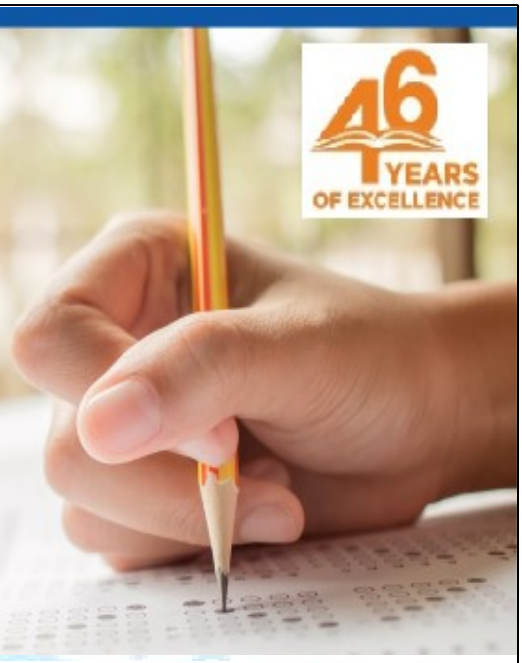
NARAYANA IIT ACADEMY INDIA

JEE ADVANCED GRAND TEST SERIES



46 YEARS OF EXCELLENCE

NARAYANA IIT - JEE ACADEMY - INDIA

Sec: OSR.IIT _*CO-SC(MODEL-A&APEX) Date: 23-04-26
Time: 3 Hrs Max. Marks: 180

23-04-26_OSR.IIT_*CO-SC(MODEL-A&APEX)_JEE ADV_GTA-8(P2)_SYLLABUS

Rankers Academy JEE

MATHEMATICS: **TOTAL SYLLABUS**

PHYSICS: **TOTAL SYLLABUS**

CHEMISTRY: **TOTAL SYLLABUS**

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TIME:3hrs **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 4)	Questions with **Single Correct** Options	+3	-1	4	12
Sec – II (Q.N : 5 – 8)	Questions with **Multiple Correct** Choice (partial marking scheme) (+1,0)	+4	-2	4	16
Sec – III (Q.N : 9 – 16)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)	+4	0	8	32
Total				**16**	**60**

PHYSICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 17 – 20)	Questions with **Single Correct** Options	+3	-1	4	12
Sec – II (Q.N : 21 – 24)	Questions with **Multiple Correct** Choice (partial marking scheme) (+1,0)	+4	-2	4	16
Sec – III (Q.N : 25 – 32)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)	+4	0	8	32
Total				**16**	**60**

CHEMISTRY

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 33 – 36)	Questions with **Single Correct** Options	+3	-1	4	12
Sec – II (Q.N : 37 – 40)	Questions with **Multiple Correct** Choice (partial marking scheme) (+1,0)	+4	-2	4	16
Sec – III (Q.N : 41 – 48)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)	+4	0	8	32
Total				**16**	**60**

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MATHEMATICS **MAX.MARKS: 60**

SECTION–I (Maximum Marks: 12)

This section contains **FOUR (04)** questions.
Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer.
For each question, choose the correct option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If only the correct option is chosen.
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

1. Let $f(x)=\left(\frac{e^{x\ln\left(2^x-1\right)}-\left(2^x-1\right)^x\sin x}{e^{x\ln x}}\right)^{\frac{1}{x}}$. Then right hand limit of $f(x)$ at $x=0$ is equal to :

A) $\ln 2$ B) $\frac{\ln 2}{e}$ C) $e^{\ln 2}$ D) does not exist

2. A function $g(\theta)=\int_0^{\sin^2\theta} f(x)dx+\int_0^{\cos^2\theta} f(x)dx$ is defined in the interval $\left(\frac{-\pi}{2},\frac{\pi}{2}\right)$ where $f(x)$ is an increasing function, then $g(\theta)$ is increasing in the interval.

A) $\left(\frac{-\pi}{2},0\right)$ B) $\left(\frac{-\pi}{2},\frac{\pi}{4}\right)$ C) $\left(0,\frac{\pi}{4}\right)$ D) $\left(\frac{-\pi}{4},0\right)$

3. In experiment I , two different colored unbiased coins are tossed. Two subsets P and Q of two elements each are randomly chosen from sample space S one after another with replacement. In experiment II, the same two coins are tossed. If the outcomes obtained match with any of the elements of sets P or Q then experiment is declared successful. Probability that experiment is not successful, is

A) $\frac{1}{2}$ B) $\frac{1}{6}$ C) $\frac{1}{4}$ D) $\frac{1}{3}$

4. If $\hat{u}$ and $\hat{v}$ are two non-collinear unit vectors such that $\left|\hat{u}+\hat{v}+2\hat{u}\times\hat{v}\right|=2$, then $\frac{\left|\hat{u}-\hat{v}\right|}{\left|\hat{u}\times\hat{v}\right|}$ is equal to

A) 1 B) 2 C) 3 D) 4

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SECTION - II (Maximum Marks : 16)

This section contains **FOUR (04)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

5. Consider $p = x\sin\alpha + \frac{x^2}{2!}\sin 2\alpha + \frac{x^3}{3!}\sin 3\alpha + \ldots\ldots\infty$ and

$q = 1 + x\cos\alpha + \frac{x^2}{2!}\cos 2\alpha + \frac{x^3}{3!}\cos 3\alpha + \ldots\ldots\infty$ where $x \in R, \alpha \in R, \alpha$ is constant and x varies then :

A) $\left(\frac{dp}{dx}\right)^2 + \left(\frac{dq}{dx}\right)^2 = p^2 + q^2$

B) $\left(\frac{dp}{dx}\right)^2 + \left(\frac{dq}{dx}\right)^2 = (p^2 + q^2)\sin^2\alpha$

C) If $\int_0^1 (p^2 + q^2)dx = f(\alpha)$ then $f(0) = \frac{e^2 - 1}{2}$

D) If $\int_0^1 (p^2 + q^2)dx = f(\alpha)$ then $f(0) = \frac{e^2 + 1}{2}$

6. If $\int \frac{dx}{(\sin x + 4)(\sin x - 1)} = \frac{A}{\tan\frac{x}{2} - 1} + B\tan^{-1}\left(\frac{4\tan\frac{x}{2} + 1}{\sqrt{15}}\right) + C$ then : (where C is constant of integration)

A) $A = \frac{2}{5}$ B) $B = \frac{-2}{5\sqrt{15}}$ C) $A = \frac{-2}{5}$ D) $B = 1$

7. If n is the smallest positive integer such that the polynomial $x^4 - nx + 63$ can be written as a product of a linear and a cubic polynomial with integer coefficients, then

A) Number of divisors of n is 10

B) Number of solutions of $ab = n$ is $10(a, b \in N)$

C) Number of odd divisors of n is 2

D) $\int_0^n \frac{e^x}{e^{[x]}}dx = 100(e - 1)$

(where [.] denotes greatest integer function)

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8. If $\sum_{i>j\geq 0}{}^{n+1}C_i\,{}^{n}C_j$ then

A) $f(2)=16$ B) $f(5)=1001$ C) $f(6)=4096$ D) $f(1)=4$

SECTION-III (Maximum Marks : 32)

This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**
For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) designated to enter the answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +3 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

9. Let $f(x)$ be a differentiable function on R defined by $f(x)=5-(x+1)^2$ and A be the point of intersection where the tangent line drawn to graph of $y=f(x)$ at the point $P(x,f(x))$ intersect with the x -axis and B be the intersection point where the tangent line intersect with y-axis. If $S(x)$ denotes the area of ΔOAB where O is the origin then the minimum value of $S(x)$ in the interval $(0,1)$ is equal to $\frac{p}{q}$ where p,q are possible numerically smallest integers. Find the value of $\frac{10q-p}{2}$

10. Let $f(x)$ is a differentiable function such that $f'(x)=f(x)+\int_0^2 f(x)dx$. If $f(0)=\frac{4-e^2}{3}$ then find number of solutions of $x+f(x)=0$.

11. If $ab^2c^3, a^2b^3c^4, a^3b^4c^5$ are in A.P. where $a>0,b>0,c>0$ then find the minimum value of $a+b+c$.

12. Lines $y=3x+4$ and $y=mx+1$ intersect parabola $x^2+x-y+1=0$ at four distinct concyclic points. If possible values of m are $m_i, i\in N$ the find $\Sigma m_i{}^2$

13. Find number of 6 digit natural numbers such that the sum of their digits is 10 and each of the digits $0,1,2,3$ occurs at least once in them.

14. If $x^2+ay^2+4xy+(2b+1)x+30y+50=0$ represents pair of parallel lines then find square of distance between them.

15. A and B are invertible square matrices of order 3 such that $|B|=2$ and $A^T|A|B=A|B|B^T$ then find value of $\left|AB^{-1}\text{adj}(A^TB)^{-1}\right|$

16. Let $S(x_1,0)$ for $x_1>0$ is focus of ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$. Suppose a parabola whose vertex is $V(x_1\sqrt{5},0)$ touches the ellipse at points A and B in I and IV quadrants respectively. Axis of parabola is x -axis. If tangent to the ellipse E at A and B meet at C then area of ΔABC is :

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PHYSICS **MAX.MARKS: 60**

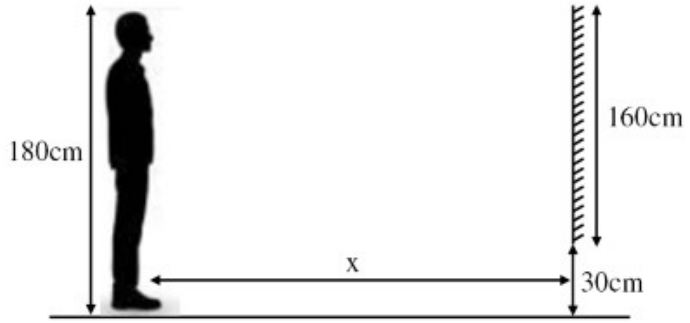
SECTION–I (Maximum Marks: 12)

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Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

17. The height of the sea level under the influence of the tide in the city of Mumbai during a full moon can be modeled as $A\sin(\omega t)$, where A is the amplitude and ω is the frequency, such that the tide occurs every 12 hours. The sailors of a ship would like to know whether they will be able to dock when they arrive at the port. The problem is that their ship can anchor at the quay only when the water level has fallen no more than $A/3$ below the maximum level and the sailors do not know at what time the high tide occurs. What is the probability that they arrive at a time when they can anchor at the quay?

A) $\frac{1}{2}-\frac{1}{\pi}\sin^{-}\left(\frac{2}{3}\right)$ B) $1-\frac{1}{2\pi}\sin^{-}\left(\frac{2}{3}\right)$ C) $1-\frac{2}{\pi}\sin^{-}\left(\frac{1}{3}\right)$ D) $\frac{1}{2}-\frac{1}{\pi}\sin^{-}\left(\frac{1}{3}\right)$

18. A person standing upright looks into a vertical plane mirror hanging on a wall. The height of the person's eyes above the floor is 180 cm , the height of the mirror is 160 cm , and the mirror is hung at a distance of 30 cm from the floor. At what distance from the mirror should the person stand in order to see themselves at full height?



A) not less than 120 cm
B) not more than 160 cm
C) not more than 240 cm
D) at any distance

19. As a result of collisions of counter-propagating beams containing lead nuclei in the CERN accelerator, among other things gold nuclei are produced. Let us assume that we have mastered a technology that allows the following reaction to be carried out on an industrial scale: ${}^{208}_{82}Pb^{3+}+{}^{208}_{82}Pb^{3+}\rightarrow 2.{}^{197}_{79}Au+6p+16n$ where ${}^{208}_{82}Pb^{3+}$ is a lead ion (a lead atom with three electrons removed), ${}^{197}_{79}Au$ is a gold atom, p is a proton, and n is a neutron. The masses of the elements involved are :
$M_{Pb^{3+}}=207.97500u, M_{Au}=196.96657u, M_p=1.007276u$, $M_n=1.008665u$, where u is

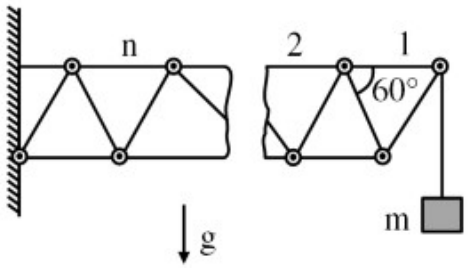
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the atomic mass unit. Assume the price of electrical energy is 1.5Rs./kWh and neglect the influence of the presence of electrons on the energy balance considered. The cost K of the energy required to produce 1 g of gold in this process satisfies:

A) $K \le 100Rs$ B) $100Rs < K \le 1000Rs$.

C) $10000Rs < K \le 20,000Rs$ D) $20,000Rs < K \le 30,000Rs$

20. A load of mass m is suspended from a system of identical massless rods connected by hinges, as shown in the figure. Determine the force that stretches the nth upper horizontal rod.



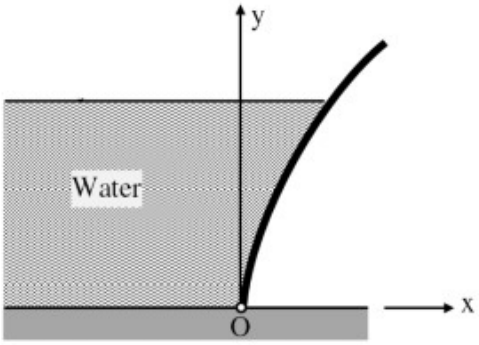
A) $\frac{mg}{\sqrt{3}}$ B) $\frac{nmg}{\sqrt{3}}$ C) $\frac{(2n-1)mg}{\sqrt{3}}$ D) $\frac{2nmg}{\sqrt{3}}$

SECTION - II (Maximum Marks : 16)

This section contains **FOUR (04)** questions.
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For each question, choose the correct option(s) to answer the question.
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Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

21. The cross-section of a gate is shown in figure. Its dimension normal to the plane of the paper is 8 m, and its shape is such that $x = 0.2y^2$. The gate is pivoted about O. The water depth is 2.5 m. Neglect atmospheric pressure. ($\rho_w = 10^3$ kg / m^3)



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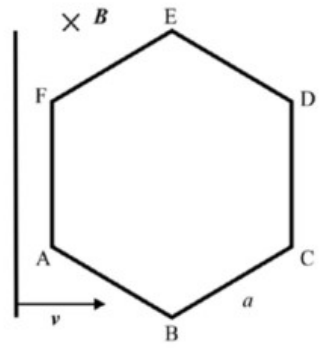
A) The horizontal component of force on the gate due to water is 250 kN .

B) The vertical component of force on the gate due to water is 83.33 kN .

C) The torque of the force on the gate about the pivot is $239\text{kN}-\text{m}$.

D) The torque of the force on the gate about the pivot is $480\text{kN}-\text{m}$.

22. A horizontal conducting loop made of wire of resistance per unit length r in the shape of a regular hexagon with side length a is placed in a homogeneous magnetic field with magnetic induction vector B perpendicular to the plane of the loop, see figure. In the plane of the loop, a straight conductor of length $\ell > 2a$ and resistance per unit length r moves parallel to the side AF of the hexagon. Its motion is uniform, and the velocity vector v is perpendicular to side AF .



A) The force F required to move the rod at a constant velocity v as a function of displacement x from AF is $\dfrac{vB^2}{2r}\dfrac{6a\left(a+\dfrac{2}{\sqrt{3}}x\right)^2}{11a^2+\dfrac{28}{\sqrt{3}}ax-\dfrac{16}{3}x^2}$.

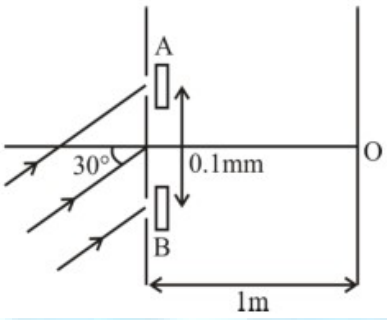
B) The potential difference between points A and F as a function of displacement x from AF is $avB\dfrac{5a^2+\dfrac{6}{\sqrt{3}}ax-\dfrac{8}{3}x^2}{11a^2+\dfrac{28}{\sqrt{3}}ax-\dfrac{16}{3}x^2}$.

C) The potential difference between points D and C as a function of displacement x from AF is $avB\dfrac{a^2+\dfrac{6}{\sqrt{3}}ax+\dfrac{8}{3}x^2}{11a^2+\dfrac{28}{\sqrt{3}}ax-\dfrac{16}{3}x^2}$.

D) The power loss in the system as a function of displacement x from AF is $\dfrac{v^2B^2}{r}\dfrac{6a\left(a+\dfrac{2}{\sqrt{3}}x\right)}{11a^2+\dfrac{28}{\sqrt{3}}ax-\dfrac{16}{3}x^2}$.

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23. In a YDSE a parallel beam of light of wavelength 6000Å is incident on slits at angle of incidence 30° as shown in figure. A and B are two thin transparent films each of refractive index 1.5. Thickness of A is $20.4\mu\ m$. Light coming through A and B have intensities I and 4I respectively on the screen. Intensity at point O which is symmetric relative to the slits is 3 I . The central maxima is above O . Choose the correct option(s).



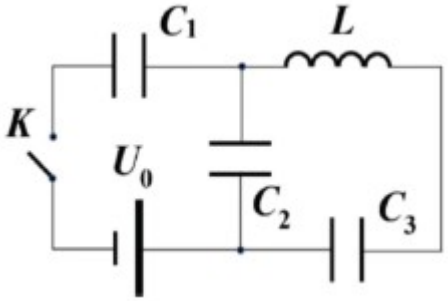
A) The thickness of B can be 240μ m

B) Fringe width is 6 mm

C) Maximum intensity on the screen is 9I

D) Minimum intensity on the screen is I

24. The electrical circuit shown schematically in the figure consists of three capacitors with capacities C_1, C_2 and C_3, a coil of inductance L and a source of constant voltage U_0. At the initial moment of time, the capacitors are not charged, and the current in the coil is zero. The switch K is closed. Assume that the resistance of the connecting wires is small. (Take : $C_1 = C_2 = C_3 = C$.)



A) The maximum current through the coil will be $\sqrt{\frac{C}{6L}}U_0$

B) The maximum current through the coil will be $\sqrt{\frac{C}{3L}}U_0$

C) The maximum voltage across the capacitor C_2 is $\frac{U_0}{3}$

D) The minimum voltage across the capacitor C_2 is $\frac{U_0}{6}$

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SECTION-III (Maximum Marks : 32)

This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**
For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the **second decimal place;** e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) designated to enter the answer.
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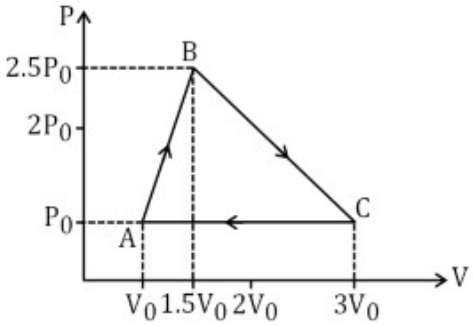
25. In an experiment to determine the charge-to-mass ratio e/m of the electron using a cylindrical diode, the following equation is derived $\frac{e}{m}=\frac{8\,V}{r^2\,B^2}$, where V is the potential difference across the diode of radius r at the critical magnetic field B . The latter is supplied by passing a current I through a solenoid of diameter D and length L , and is given by $B=\mu_0 nI$, where n is the number of turns per unit length. The experimental quantities are determined as follows :

$n=3920\ m^{-1};r=(3.4\pm0.1)\times10^{-3}\ m;I=1.92\pm0.02\ A;V=20\pm1\ V$

The error (in %) in measurement of ratio $\left(\frac{e}{m}\right)$ is :-

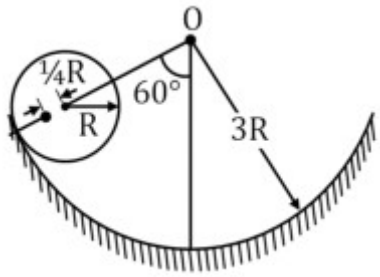
26. A reversible heat engine in a satellite operates between a hot reservoir at T_1 and a radiating panel at T_2. Radiation from the panel is proportional to its area and to $T_2^{\,4}$. Determine the minimum area (in m^2) of the panel for an output of 1 kW if T_1 is 1000 K . The panels behave like a black body.

27. A certain designer built a complicated heat engine whose working substance is n moles of a monatomic ideal gas, operating in a cycle that forms a triangle ABC on the P – V diagram, with vertices at the points (V_0,P_0), $(3/2\ V_0,5/2P_0)$, and $(3\ V_0,P_0)$. Calculate the efficiency (in %) of this engine.

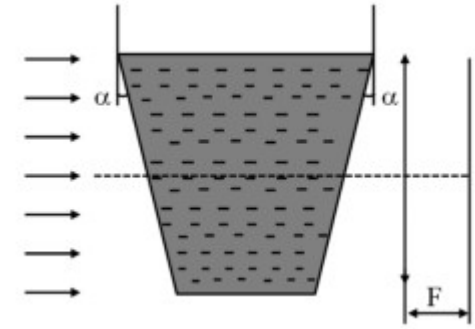


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28. A particle of mass m is embedded at a distance $R/4$ from the center of a massless circular disk of radius R which can roll without slipping on the inside surface of a fixed circular cylinder of radius $3R$. The disk is released with zero velocity from the position shown and rolls because of gravity, all motion taking place in the same vertical plane. If the reaction force acting on the disk at the point of contact when it is at its lowest position is αmg. The value of α is :-

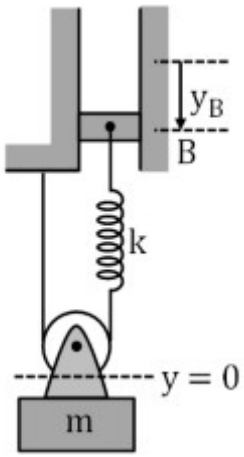


29. A glass trapezoidal vessel (see figure) with a small angle $\alpha = 6^\circ$ is filled with water with refractive index $n = 1.33$. A parallel beam of light is incident on the vessel. Behind the vessel there is a converging lens with focal length $F = 50$ cm. On a screen placed in the focal plane of the lens, a bright point is observed. By how much will this point (in cm) shift on the screen if the vessel is removed?

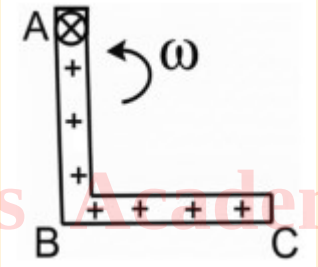


30. The equilibrium position of the mass m occurs where $y = 0$ and $y_B = 0$. When the attachment B is given a steady vertical motion $y_B = b\sin\omega t$, the mass m will acquire a steady vertical oscillation. Find the angular frequency ω for which the oscillations of m tend to become excessively large. The stiffness of the spring is k, and the mass and friction of the pulley are negligible. (Take: $k = 10$ N/m, $m = 0.1$ kg.)

For More Join: @IITJEE_Advanced



31. A uniform thin L -shaped rod of mass m carries charge q uniformly over its length. Rod is rotating about axis through end A as shown. If $AB = BC = L$ then magnetic dipole moment of rod is $\frac{\alpha q \omega L^2}{2}$. Find α.



32. At a particular instant a source of sound of frequency 1000 Hz is at $(2\text{ m}, 1\text{ m})$ and an observer is at $(5\text{ m}, 5\text{ m})$. The velocity of source and observer are $30(\hat{i}+\hat{j})\text{ m/s}$ and $15(\hat{i}+\sqrt{3}\hat{j})\text{ m/s}$ respectively at this instant. The velocity of sound in air is 330 m/s. If observer receives frequency f Hz then the value of f is :-

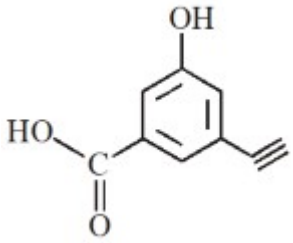
For More Join: @IITJEE_Advanced

CHEMISTRY **MAX.MARKS: 60**

SECTION–I (Maximum Marks: 12)

This section contains **FOUR (04)** questions.
Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer.
For each question, choose the correct option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If only the correct option is chosen.
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

33. Incorrect statement about the given compound



A) liberates H_2 on treatment with Na B) gives positive test with neutral $FeCl_3$
C) reacts with Tollen's reagent D) gives 2,4-DNP test

34. A volume V of a gas at a temperature T_1 and a pressure p is enclosed in a sphere. It is connected to another sphere of volume $\left(\frac{V}{2}\right)$ by a tube and stopcock. The second sphere is initially evacuated and the stopcock is closed. If the stopcock is opened, the temperature of the gas in the second sphere becomes T_2. The first sphere is maintained at a temperature T_1. What is the final pressure p' within the apparatus?

A) $\frac{2pT_2}{2T_2+T_1}$ B) $\frac{2pT_2}{T_2+2T_1}$ C) $\frac{pT_2}{2T_2+T_1}$ D) $\frac{pT_2}{T_2+T_1}$

35. Suppose, the wave function of orbital of a H-like atom is

$$\psi_{2s}=\frac{1}{4\sqrt{2\pi}}Z^{3/2}(2-Zr)e^{-Zr/2}\ (r \text{ is in Å})$$

Which of the following will be radius of nodal surface for He^+ ion?

A) 1Å B) 2Å C) 2.5Å D) 4Å

36. When aqueous solution of Zinc(II) nitrate is gently warmed with sodium hydroxide and an alloy $(45\%Al, 5\%Zn, 50\%Cu)$ a gas is evolved, that gas can be detected by

(I) Its action upon moist red litmus paper
(II) Its action upon moist blue litmus paper
(III) Its action upon mercury(I) nitrate on moist paper
(IV) Its action in lime water
(V) Tannic acid - silver nitrate test

A) II, IV and V B) I, III and V C) Only III and V D) I, III & IV

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SECTION - II (Maximum Marks : 16)

This section contains **FOUR (04)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

37. When an organic compound is fused with sodium and the extract in treated with neutral $FeCl_3$ a blood red colour is formed. Addition of ____ decrease the intensity of blood red colour :

Select correct option to fill the blank space.

A) Aqueous $HgCl_2$ B) Aqueous $H_2C_2O_4$ (oxalic acid)
C) 0.002 M potassium thiocyanate D) Aqueous KF

38. Biuret test is given by

(A) Proteins (B) Carbohydrates (C) Polypeptides (D) Heating urea

39. Two moles of an ideal gas $(C_{v,m} = 1.5R)$ is subjected to the following changes in states:

$$\underset{(500K,\ 5bar)}{(A)} \xrightarrow[\text{Expansion}]{\text{Reversible Isothermal}} (B) \xrightarrow{\text{Isochoric Cooling}} \underset{(250K,\ 1bar)}{(C)} \xrightarrow[\text{compression}]{\text{Single statge adiabatic}} \underset{(3\ bar)}{(D)}$$

The correct statement(s) is/are

A) The pressure at B is 2.0 bar. B) The temperature at D is 450 K .
C) $\Delta H_{CD} = 1000R$ D) $\Delta U_{BC} = 375R$

40. If a hydrogen molecule absorbs a photon having energy exactly equal to the energy difference in between antibonding is σ^*_{1s}MO and bonding σ_{1s}MO of H_2 molecule, what happens?

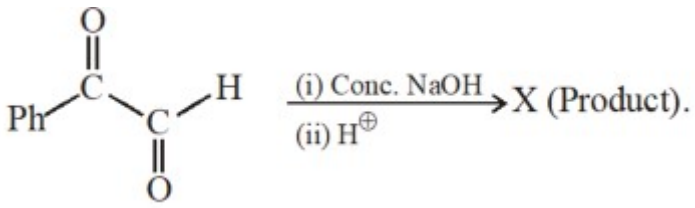
A) The H_2 molecule will ionize
B) The H_2 molecule will get stabilized
C) The bond between molecule breaks and dissociated atoms have some kinetic energy
D) The bond between molecule breaks and dissociated atoms have zero kinetic energy

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SECTION-III (Maximum Marks : 32)

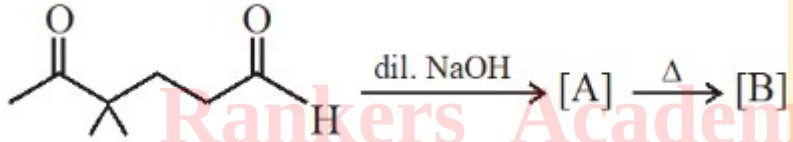
This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**
For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the **second decimal place;** e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) designated to enter the answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +3 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

41.



One mole of product X formed is treated with excess PCl_5 to give product Y.Y is then treated with excess CH_3MgBr and hydrolyzed with H^+ / H_2O. What is the total mass of CH_3MgBr consumed ?
(Atomic mass $Mg = 24, Br = 80$)

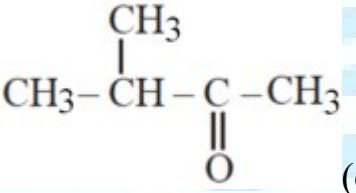
42.

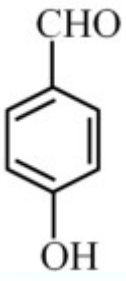


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Total number of H -atoms in the ring of both the compounds A and B are ____ .

43. The number of compounds which does not give Cannizzaro reaction





(a) Chloral (b) (c) (d) HCHO (e) PhCHO

44. How many statement are correct for the reaction : $A \rightarrow$ Products (order = n), the theoretical time of completion of reaction is
(i) Infinite for all the value of n
(ii) Infinite for $n \geq 1$
(iii) $\frac{[A_0]^{1-n}}{K(1-n)}$ for $n \neq 1$
(iv) $\frac{[A_0]^{1-n}}{K(1-n)}$ for $n < 1$

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45. Equimolar mixture of two gases, A_2 and B_2, is taken in a rigid vessel at constant temperature 300 K . The gases achieve equilibrium as:

$A_2(g) \rightleftharpoons 2A(g), K_P = x\text{atm}$

$B_2(g) \rightleftharpoons 2B(g), K_P = y\text{atm}$

$A_2(g) + B_2(g) \rightleftharpoons 2AB(g), K_P = 2$

If the initial pressure in the container was 2 atm and the final pressure at equilibrium is 2.75 atm in which the partial pressure of $AB(g)$ is 0.5 atm , the value of $y:x$ is {$y:x$ is INTEGER value }

46. $I_{2(s)} / I^-_{(aq)}(0.1M)$ half cell is connected to a $H^+_{(aq)} / H_2(1\text{bar}) / Pt$ half cell and emf is found to be 0.7714 V . If $E^0_{I_2/I^-} = 0.535$ V then find pH of H^+ / H_2 half cell.

$\left[\frac{2.303RT}{F} = 0.0591\right]$ (Report answer to nearest integer)

47. A coordination compound X has the molecular formula : $M(en)_2Cl_3$ Where en = ethylenediamine [bidentate] The Compound shows the following properties.

(M = Metal of 3d series have atomic number higher than 21)

(i) when dissolved in water, compound X produces 2 ions in solution.

(ii) The compound is found to have no unpaired electron.

(iii) The compound exhibits both geometrical and optical isomerism.

(iv) On treatment with excess aqueous $AgNO_3$, one mole of compound gives one mole of white precipitate.

Find the value of : $A + B + C + D + E + F$ where :

A : Atomic number of M

B : Number of t_{2g} electrons on metal ion in complex

C : Total number of geometrical isomers of X

D : Total number of optically active isomers of X

E : Number of chloride ions present outside Coordination Sphere

F : Missing value in calculation of crystal field stabilization energy of $X = -(\ldots\ldots\ldots)\Delta_0$ (by neglecting pairing energy)

48.

$$\underset{(620\text{ gm})}{P_4} \xrightarrow[-\text{Gas(X)}]{(\text{Excess KOH Aq.})} \underset{(\text{Salt})}{[A]} \xrightarrow{(\text{Excess dil } H^+)} \text{Acid B} \xrightarrow{(\text{Excess}AgNO_3\text{Aq.})} \text{Precipitate (C)}$$

(Given atomic weight of $P = 31, H = 1, O = 16, H = 1, N = 14$ $K = 39, Ag = 108$)

Calculate (Mass of gas (X) + Mass of Precipitate (C)) in gm.

Assuming 100% yield of each step.

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